Progress Report April 13th, 2026

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Project: Influence of Decade on Song Lengths

This week we have done a blocking Anova Analysis of 270 songs from apple music. The songs are sample in blocks of 45 from six decade {1970, 1980, 1990, 2000, 2010, 2020}. The 45 songs are selected by three sets of top 15 billboard lists for their respective year, from three different years from each decade. Each Song is classified in 12 genres { Afrobeats, Country, Disco, EDM, Funk, Hip-Hop, Latin, Pop, R&B, Reggae, Rock, Soul}. Data Presented in Figures 1:6.

	A	B	C	D	E	F	G	H	I
1	DECADE	YEAR	GENRE	ARTIST	SONG TITLE	RANK	LENGTH (MM:SS)	SEC	PLATFORM
2	1970	1978	Disco	Andy Gibb	Shadow Dancing	1	4:36	276	Apple Music
3	1970	1978	Disco	Bee Gees	Night Fever	2	3:32	212	Apple Music
4	1970	1978	Pop	Debby Boone	You Light Up My Life	3	3:36	216	Apple Music
5	1970	1978	Disco	Bee Gees	Stayin' Alive	4	1:33	93	Apple Music
6	1970	1978	Pop	Exile	Kiss You All Over	5	4:57	297	Apple Music
7	1970	1978	Pop	Bee Gees	How Deep Is Your Love	6	4:05	245	Apple Music
8	1970	1978	Rock	Player	Baby Come Back	7	4:16	256	Apple Music
9	1970	1978	Disco	Andy Gibb	(Love Is) Thicker Than Water	8	4:16	256	Apple Music
10	1970	1978	Disco	A Taste of Honey	Boogie Oogie Oogie	9	3:44	224	Apple Music
11	1970	1978	Soul	Commoderes	Three Times a Lady	10	3:40	220	Apple Music
12	1970	1978	Disco	Frankie Valli	Grease	11	3:27	207	Apple Music
13	1970	1978	Pop	Paul Davis	I Go Crazy	12	3:52	232	Apple Music
14	1970	1978	Disco	John Travolta & Olivia Newton-John	You're the One That I want	13	2:49	169	Apple Music
15	1970	1978	Pop	Samantha Sang	Emotion	14	3:57	237	Apple Music
16	1970	1978	Rock	Eric Clapton	Lay Down Sally	15	3:56	236	Apple Music
17	1970	1970	Pop	Simon & Garfunkel	"Bridge Over Troubled Water"	1	4:56	296	Apple Music
18	1970	1970	Pop	The Carpenters	"(They Long to Be) Close to You"	2	4:36	276	Apple Music
19	1970	1970	Rock	The Guess Who	"American Woman"	3	5:08	308	Apple Music
20	1970	1970	Pop	B.J. Thomas	"Raindrops Keep Fallin' on My Head"	4	3:04	184	Apple Music
21	1970	1970	Soul	Edwin Starr	"War"	5	3:50	230	Apple Music
22	1970	1970	Soul	Diana Ross	"Ain't No Mountain High Enough"	6	6:18	378	Apple Music
23	1970	1970	Soul	The Jackson 5	"I'll Be There"	7	3:59	239	Apple Music
24	1970	1970	Rock	Rare Earth	"Get Ready"	8	2:54	174	Apple Music
25	1970	1970	Rock	The Beatles	"Let It Be"	9	4:03	243	Apple Music
26	1970	1970	Soul	Freda Payne	"Band of Gold"	10	2:57	177	Apple Music
27	1970	1970	Rock	Three Dog Night	"Mama Told Me (Not to Come)"	11	3:19	199	Apple Music
28	1970	1970	Pop	Ray Stevens	"Everything Is Beautiful"	12	3:33	213	Apple Music
29	1970	1970	Rock	Bread	"Make It with You"	13	3:12	192	Apple Music
30	1970	1970	Pop	Vanity Fare	"Hitchin' a Ride"	14	2:58	178	Apple Music
31	1970	1970	Soul	The Jackson 5	"ABC"	15	2:58	178	Apple Music
32	1970	1972	Soul	Roberta Flack	"The First Time Ever I Saw Your Face"	1	4:20	260	Apple Music
33	1970	1972	Pop	Gilbert O'Sullivan	"Alone Again (Naturally)"	2	3:42	222	Apple Music
34	1970	1972	Rock	Don McLean	"American Pie"	3	8:34	514	Apple Music
35	1970	1972	Pop	Harry Nilsson	"Without You"	4	3:22	202	Apple Music
36	1970	1972	Pop	Sammy Davis Jr.	"The Candy Man"	5	3:13	193	Apple Music
37	1970	1972	Funk	Joe Tex	"I Gotcha"	6	2:29	149	Apple Music
38	1970	1972	Soul	Bill Withers	"Lean on Me"	7	4:18	258	Apple Music
39	1970	1972	Country	Mac Davis	"Baby, Don't Get Hooked on Me"	8	3:04	184	Apple Music
40	1970	1972	Pop	Melanie	"Brand New Key"	9	2:25	145	Apple Music
41	1970	1972	Pop	Wayne Newton	"Daddy Don't You Walk So Fast"	10	3:19	199	Apple Music
42	1970	1972	Soul	Al Green	"Let's Stay Together"	11	3:19	199	Apple Music
43	1970	1972	Pop	Looking Glass	"Brandy (You're a Fine Girl)"	12	3:09	189	Apple Music
44	1970	1972	Soul	The Chi-Lites	"Oh Girl"	13	3:16	196	Apple Music
45	1970	1972	Pop	Gallery	"Nice to Be with You"	14	3:05	185	Apple Music
46	1970	1972	Rock	Chuck Berry	"My Ding-a-Ling"	15	4:40	280	Apple Music

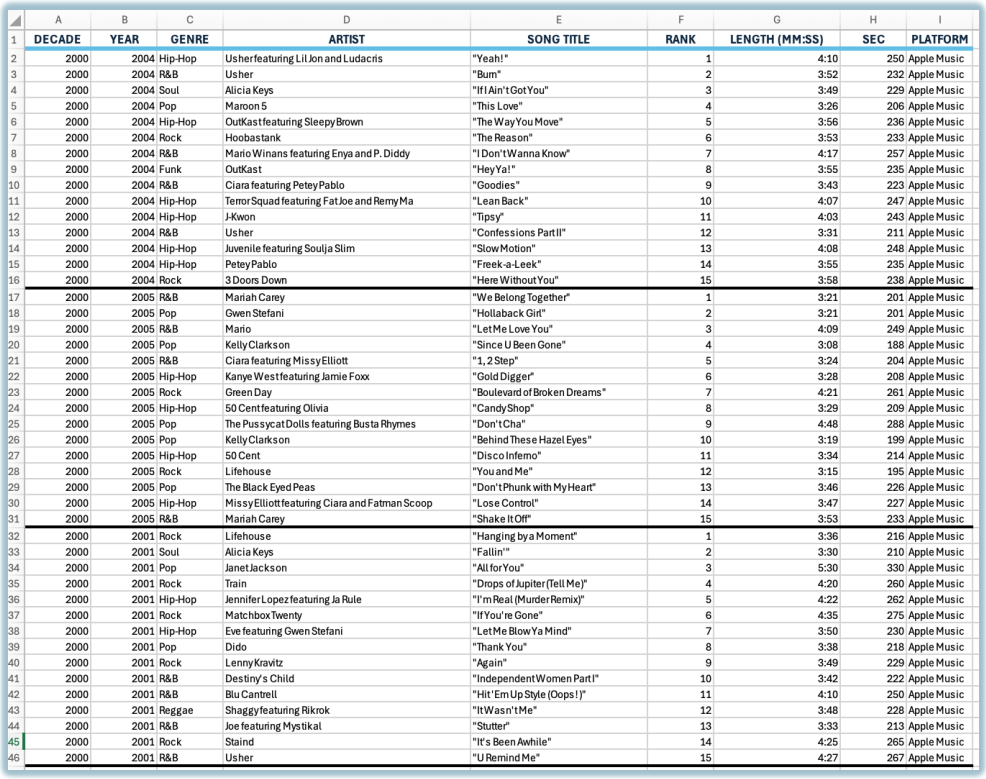
Figure 1: 1970s Decade Data Block

	A	B	C	D	E	F	G	H	I
1	Decade	YEAR	GENRE	ARTIST	SONG TITLE	RANK	LENGTH (MM:SS)	SEC	PLATFORM
2	1980	1986	Soul	Dionne and Friends (Dionne Warwick, Gladys	"That's What Friends Are For"	1	4:16	256	Apple Music
3	1980	1986	Pop	Lionel Richie	"Say You, Say Me"	2	4:03	243	Apple Music
4	1980	1986	R&B	Klymaxx	"I Miss You"	3	5:32	332	Apple Music
5	1980	1986	R&B	Patti LaBelle and Michael McDonald	"On My Own"	4	4:43	283	Apple Music
6	1980	1986	Rock	Mr. Mister	"Broken Wings"	5	4:44	284	Apple Music
7	1980	1986	Pop	Whitney Houston	"How Will I Know"	6	4:30	270	Apple Music
8	1980	1986	R&B	Eddie Murphy	"Party All the Time"	7	4:13	253	Apple Music
9	1980	1986	Rock	Survivor	"Burning Heart"	8	3:50	230	Apple Music
10	1980	1986	Rock	Mr. Mister	"Kyrie"	9	4:17	257	Apple Music
11	1980	1986	Rock	Robert Palmer	"Addicted to Love"	10	6:04	364	Apple Music
12	1980	1986	Soul	Whitney Houston	"The Greatest Love of All"	11	4:57	297	Apple Music
13	1980	1986	R&B	Atlantic Starr	"Secret Lovers"	12	5:31	331	Apple Music
14	1980	1986	Pop	Gloria Loring and Carl Anderson	"Friends and Lovers"	13	3:47	227	Apple Music
15	1980	1986	Pop	Peter Cetera	"Glory of Love"	14	4:18	258	Apple Music
16	1980	1986	Disco	Pet Shop Boys	"West End Girls"	15	4:46	286	Apple Music
17	1980	1984	Funk	Prince	"When Doves Cry"	1	5:54	354	Apple Music
18	1980	1984	Rock	Tina Turner	"What's Love Got to Do with It"	2	3:52	232	Apple Music
19	1980	1984	Pop	Paul McCartney and Michael Jackson	"Say, Say, Say"	3	2:55	175	Apple Music
20	1980	1984	Rock	Kenny Loggins	"Footloose"	4	3:41	221	Apple Music
21	1980	1984	Pop	Phil Collins	"Against All Odds (Take a Look at Me Now)"	5	3:26	206	Apple Music
22	1980	1984	Rock	Van Halen	"Jump"	6	4:02	242	Apple Music
23	1980	1984	Pop	Lionel Richie	"Hello"	7	4:09	249	Apple Music
24	1980	1984	Rock	Yes	"Owner of a Lonely Heart"	8	4:30	270	Apple Music
25	1980	1984	Pop	Ray Parker Jr.	"Ghostbusters"	9	4:05	245	Apple Music
26	1980	1984	Pop	Culture Club	"Karma Chameleon"	10	4:13	253	Apple Music
27	1980	1984	Rock	John Waite	"Missing You"	11	4:30	270	Apple Music
28	1980	1984	Disco	Lionel Richie	"All Night Long (All Night)"	12	6:22	382	Apple Music
29	1980	1984	Disco	Deniece Williams	"Let's Hear It for the Boy"	13	4:22	262	Apple Music
30	1980	1984	Rock	Bruce Springsteen	"Dancing in the Dark"	14	4:05	245	Apple Music
31	1980	1984	Pop	Cyndi Lauper	"Girls Just Want to Have Fun"	15	3:58	238	Apple Music
32	1980	1988	Pop	George Michael	"Faith"	1	3:16	196	Apple Music
33	1980	1988	Rock	INXS	"Need You Tonight"	2	3:01	181	Apple Music
34	1980	1988	Rock	George Harrison	"Got My Mind Set on You"	3	3:55	235	Apple Music
35	1980	1988	Pop	Rick Astley	"Never Gonna Give You Up"	4	3:34	214	Apple Music
36	1980	1988	Rock	Guns N' Roses	"Sweet Child o' Mine"	5	5:56	356	Apple Music
37	1980	1988	Pop	Whitney Houston	"So Emotional"	6	4:37	277	Apple Music
38	1980	1988	Pop	Belinda Carlisle	"Heaven Is a Place on Earth"	7	4:07	247	Apple Music
39	1980	1988	Pop	Tiffany	"Could've Been"	8	3:33	213	Apple Music
40	1980	1988	Pop	Breathe	"Hands to Heaven"	9	4:20	260	Apple Music
41	1980	1988	Rock	Steve Winwood	"Roll with It"	10	5:21	321	Apple Music
42	1980	1988	Pop	George Michael	"One More Try"	11	5:51	351	Apple Music
43	1980	1988	Soul	Terence Trent D'Arby	"Wishing Well"	12	3:30	210	Apple Music
44	1980	1988	Latin	Gloria Estefan and Miami Sound Machine	"Anything for You"	13	3:42	222	Apple Music
45	1980	1988	Rock	Cheap Trick	"The Flame"	14	5:39	339	Apple Music
46	1980	1988	Disco	Billy Ocean	"Get Outta My Dreams, Get into My Car"	15	4:44	284	Apple Music

Figure 2: 1980s Decade Data Block

	A	B	C	D	E	F	G	H	I
1	DECADE	YEAR	GENRE	ARTIST	SONG TITLE	RANK	LENGTH (MM:SS)	SEC	PLATFORM
2	1990	1990	Pop	"Hold On"	Wilson Phillips	1	4:27	267	Apple Music
3	1990	1990	Pop	"It Must Have Been Love"	Roxette	2	4:20	260	Apple Music
4	1990	1990	Pop	"Nothing Compares 2 U"	Sinéad O'Connor	3	4:40	280	Apple Music
5	1990	1990	R&B	"Poison"	Bell Biv DeVoe	4	4:22	262	Apple Music
6	1990	1990	Disco	"Vogue"	Madonna	5	5:17	317	Apple Music
7	1990	1990	R&B	"Vision of Love"	Mariah Carey	6	3:29	209	Apple Music
8	1990	1990	Pop	"Another Day in Paradise"	Phil Collins	7	5:22	322	Apple Music
9	1990	1990	R&B	"Hold On"	En Vogue	8	5:04	304	Apple Music
10	1990	1990	Rock	"Cradle of Love"	Billy Idol	9	4:39	279	Apple Music
11	1990	1990	Rock	"Blaze of Glory"	Jon Bon Jovi	10	5:35	335	Apple Music
12	1990	1990	R&B	"Do Me!"	Bell Biv DeVoe	11	4:32	272	Apple Music
13	1990	1990	Pop	"How Am I Supposed to Live Without You"	Michael Bolton	12	4:14	254	Apple Music
14	1990	1990	EDM	"Pump Up the Jam"	Technotronic	13	5:20	320	Apple Music
15	1990	1990	Pop	"Opposites Attract"	Paula Abdul	14	3:50	230	Apple Music
16	1990	1990	R&B	"Escapade"	Janet Jackson	15	4:44	284	Apple Music
17	1990	1992	R&B	"End of the Road"	Boyz II Men	1	5:51	351	Apple Music
18	1990	1992	Hip-Hop	"Baby Got Back"	Sir Mix-a-Lot	2	4:23	263	Apple Music
19	1990	1992	Hip-Hop	"Jump"	Kris Kross	3	3:15	195	Apple Music
20	1990	1992	Pop	"Save the Best for Last"	Vanessa Williams	4	3:39	219	Apple Music
21	1990	1992	R&B	"Baby-Baby-Baby"	TLC	5	5:15	315	Apple Music
22	1990	1992	Rock	"Tears in Heaven"	Eric Clapton	6	4:32	272	Apple Music
23	1990	1992	R&B	"My Lovin' (You're Never Gonna Get It)"	En Vogue	7	4:42	282	Apple Music
24	1990	1992	Rock	"Under the Bridge"	Red Hot Chili Peppers	8	4:24	264	Apple Music
25	1990	1992	R&B	"All 4 Love"	Color Me Badd	9	3:31	211	Apple Music
26	1990	1992	Pop	"Just Another Day"	Jon Secada	10	5:27	327	Apple Music
27	1990	1992	Pop	"I Love Your Smile"	Shanice	11	4:18	258	Apple Music
28	1990	1992	Rock	"To Be with You"	Mr. Big	12	3:27	207	Apple Music
29	1990	1992	Disco	"I'm Too Sexy"	Right Said Fred	13	2:50	170	Apple Music
30	1990	1992	Pop	"Black or White"	Michael Jackson	14	4:17	257	Apple Music
31	1990	1992	Country	"Achy Breaky Heart"	Billy Ray Cyrus	15	3:25	205	Apple Music
32	1990	1996	Latin	"Macarena (Bayside Boys Mix)"	Los del Río	1	4:12	252	Apple Music
33	1990	1996	R&B	"One Sweet Day"	Mariah Carey and Boyz II Men	2	4:41	281	Apple Music
34	1990	1996	Pop	"Because You Loved Me"	Celine Dion	3	4:36	276	Apple Music
35	1990	1996	R&B	"Nobody Knows"	The Tony Rich Project	4	5:07	307	Apple Music
36	1990	1996	R&B	"Always Be My Baby"	Mariah Carey	5	4:18	258	Apple Music
37	1990	1996	Rock	"Give Me One Reason"	Tracy Chapman	6	4:28	268	Apple Music
38	1990	1996	Hip-Hop	"Tha Crossroads"	Bone Thugs-n-Harmony	7	3:43	223	Apple Music
39	1990	1996	Pop	"I Love You Always Forever"	Donna Lewis	8	4:00	240	Apple Music
40	1990	1996	R&B	"You're Makin' Me High" / "Let It Flow"	Toni Braxton	9	4:27	267	Apple Music
41	1990	1996	R&B	"Twisted"	Keith Sweat	10	4:13	253	Apple Music
42	1990	1996	Hip-Hop	"C'mon N' Ride It (The Train)"	Quad City DJ's	11	7:31	451	Apple Music
43	1990	1996	EDM	"Missing"	Everything but the Girl	12	4:07	247	Apple Music
44	1990	1996	Rock	"Ironic"	Alanis Morissette	13	3:48	228	Apple Music
45	1990	1996	Soul	"Exhale (Shoop Shoop)"	Whitney Houston	14	3:24	204	Apple Music
46	1990	1996	Rock	"Follow You Down" / "Til I Hear It from You"	Gin Blossoms	15	4:30	270	Apple Music

Figure 3: 1980s Decade Data Block



DECADE	YEAR	GENRE	ARTIST	SONG TITLE	RANK	LENGTH (MM:SS)	SEC	PLATFORM
2000	2004	Hip-Hop	Usher featuring Lil Jon and Ludacris	"Yeah!"	1	4:10	250	Apple Music
2000	2004	R&B	Usher	"Burn"	2	3:52	232	Apple Music
2000	2004	Soul	Alicia Keys	"If I Ain't Got You"	3	3:49	229	Apple Music
2000	2004	Pop	Maroon 5	"This Love"	4	3:26	206	Apple Music
2000	2004	Hip-Hop	OutKast featuring Sleepy Brown	"The Way You Move"	5	3:56	236	Apple Music
2000	2004	Rock	Hoobastank	"The Reason"	6	3:53	233	Apple Music
2000	2004	R&B	Mario Winans featuring Enya and P. Diddy	"I Don't Wanna Know"	7	4:17	257	Apple Music
2000	2004	Funk	OutKast	"Hey Ya!"	8	3:55	235	Apple Music
2000	2004	R&B	Ciara featuring Petey Pablo	"Goodies"	9	3:43	223	Apple Music
2000	2004	Hip-Hop	Terror Squad featuring Fat Joe and Remy Ma	"Lean Back"	10	4:07	247	Apple Music
2000	2004	Hip-Hop	J-Kwon	"Tipsy"	11	4:03	243	Apple Music
2000	2004	R&B	Usher	"Confessions Part II"	12	3:31	211	Apple Music
2000	2004	Hip-Hop	Juvenile featuring Soulja Slim	"Slow Motion"	13	4:08	248	Apple Music
2000	2004	Hip-Hop	Petey Pablo	"Freek-a-Leek"	14	3:55	235	Apple Music
2000	2004	Rock	3 Doors Down	"Here Without You"	15	3:58	238	Apple Music
2000	2005	R&B	Mariah Carey	"We Belong Together"	1	3:21	201	Apple Music
2000	2005	Pop	Gwen Stefani	"Hollaback Girl"	2	3:21	201	Apple Music
2000	2005	R&B	Mario	"Let Me Love You"	3	4:09	249	Apple Music
2000	2005	Pop	Kelly Clarkson	"Since U Been Gone"	4	3:08	188	Apple Music
2000	2005	R&B	Ciara featuring Missy Elliott	"1, 2 Step"	5	3:24	204	Apple Music
2000	2005	Hip-Hop	Kanye West featuring Jamie Foxx	"Gold Digger"	6	3:28	208	Apple Music
2000	2005	Rock	Green Day	"Boulevard of Broken Dreams"	7	4:21	261	Apple Music
2000	2005	Hip-Hop	50 Cent featuring Olivia	"Candy Shop"	8	3:29	209	Apple Music
2000	2005	Pop	The Pussycat Dolls featuring Busta Rhymes	"Don't Cha"	9	4:48	288	Apple Music
2000	2005	Pop	Kelly Clarkson	"Behind These Hazel Eyes"	10	3:19	199	Apple Music
2000	2005	Hip-Hop	50 Cent	"Disco Inferno"	11	3:34	214	Apple Music
2000	2005	Rock	Lifehouse	"You and Me"	12	3:15	195	Apple Music
2000	2005	Pop	The Black Eyed Peas	"Don't Phunk with My Heart"	13	3:46	226	Apple Music
2000	2005	Hip-Hop	Missy Elliott featuring Ciara and Fatman Scoop	"Lose Control"	14	3:47	227	Apple Music
2000	2005	R&B	Mariah Carey	"Shake It Off"	15	3:53	233	Apple Music
2000	2001	Rock	Lifehouse	"Hanging by a Moment"	1	3:36	216	Apple Music
2000	2001	Soul	Alicia Keys	"Fallin'"	2	3:30	210	Apple Music
2000	2001	Pop	Janet Jackson	"All for You"	3	5:30	330	Apple Music
2000	2001	Rock	Train	"Drops of Jupiter (Tell Me)"	4	4:20	260	Apple Music
2000	2001	Hip-Hop	Jennifer Lopez featuring Ja Rule	"I'm Real (Murder Remix)"	5	4:22	262	Apple Music
2000	2001	Rock	Matchbox Twenty	"If You're Gone"	6	4:35	275	Apple Music
2000	2001	Hip-Hop	Eve featuring Gwen Stefani	"Let Me Blow Ya Mind"	7	3:50	230	Apple Music
2000	2001	Pop	Dido	"Thank You"	8	3:38	218	Apple Music
2000	2001	Rock	Lenny Kravitz	"Again"	9	3:49	229	Apple Music
2000	2001	R&B	Destiny's Child	"Independent Women Part I"	10	3:42	222	Apple Music
2000	2001	R&B	Blu Cantrell	"Hit 'Em Up Style (Oops!)"	11	4:10	250	Apple Music
2000	2001	Reggae	Shaggy featuring Rikrok	"It Wasn't Me"	12	3:48	228	Apple Music
2000	2001	R&B	Joe featuring Mystikal	"Stutter"	13	3:33	213	Apple Music
2000	2001	Rock	Staind	"It's Been Awhile"	14	4:25	265	Apple Music
2000	2001	R&B	Usher	"U Remind Me"	15	4:27	267	Apple Music

Figure 4: 2000s Decade Data Block



2010	2011	EDM	LMFAO featuring Lauren Bennett and GoonRock	"Party Rock Anthem"	2	4:22	262	Apple Music
2010	2011	Pop	Katy Perry	"Firework"	3	3:48	228	Apple Music
2010	2011	Pop	Katy Perry featuring Kanye West	"E.T."	4	3:50	230	Apple Music
2010	2011	EDM	Pitbull featuring Ne-Yo, Afrojack and Nayer	"Give Me Everything"	5	4:12	252	Apple Music
2010	2011	Pop	Bruno Mars	"Grenade"	6	3:42	222	Apple Music
2010	2011	Soul	CeeLo Green	"Fuck You (Forget You)"	7	3:43	223	Apple Music
2010	2011	Hip-Hop	Nicki Minaj	"Super Bass"	8	3:20	200	Apple Music
2010	2011	Pop	Maroon 5 featuring Christina Aguilera	"Moves like Jagger"	9	3:21	201	Apple Music
2010	2011	Pop	The Black Eyed Peas	"Just Can't Get Enough"	10	3:39	219	Apple Music
2010	2011	EDM	Jennifer Lopez featuring Pitbull	"On the Floor"	11	4:45	285	Apple Music
2010	2011	Pop	Rihanna	"S&M"	12	4:04	244	Apple Music
2010	2011	Rock	Foster the People	"Pumped Up Kicks"	13	4:00	240	Apple Music
2010	2011	Pop	Katy Perry	"Last Friday Night (T.G.I.F.)"	14	3:51	231	Apple Music
2010	2011	Pop	Bruno Mars	"Just the Way You Are"	15	3:41	221	Apple Music
2010	2013	Hip-Hop	Macklemore & Ryan Lewis featuring Wanz	"Thrift Shop"	1	3:56	236	Apple Music
2010	2013	Funk	Robin Thicke featuring T.I. and Pharrell Williams	"Blurred Lines"	2	4:23	263	Apple Music
2010	2013	Rock	Imagine Dragons	"Radioactive"	3	3:07	187	Apple Music
2010	2013	EDM	Baauer	"Harlem Shake"	4	3:17	197	Apple Music
2010	2013	Hip-Hop	Macklemore & Ryan Lewis featuring Ray Dalton	"Can't Hold Us"	5	4:18	258	Apple Music
2010	2013	Pop	Justin Timberlake	"Mirrors"	6	8:05	485	Apple Music
2010	2013	Pop	Pink featuring Nate Ruess	"Just Give Me a Reason"	7	4:03	243	Apple Music
2010	2013	Pop	Bruno Mars	"When I Was Your Man"	8	3:31	211	Apple Music
2010	2013	Country	Florida Georgia Line featuring Nelly	"Cruise"	9	3:30	210	Apple Music
2010	2013	Pop	Katy Perry	"Roar"	10	3:44	224	Apple Music
2010	2013	Pop	Bruno Mars	"Locked Out of Heaven"	11	3:53	233	Apple Music
2010	2013	Rock	The Lumineers	"Ho Hey"	12	2:43	163	Apple Music
2010	2013	Pop	Rihanna featuring Mikky Ekko	"Stay"	13	4:01	241	Apple Music
2010	2013	Disco	Daft Punk featuring Pharrell Williams	"Get Lucky"	14	6:10	370	Apple Music
2010	2013	Pop	Lorde	"Royals"	15	3:10	190	Apple Music
2010	2019	Country	Lil Nas X featuring Billy Ray Cyrus	"Old Town Road"	1	2:37	157	Apple Music
2010	2019	Hip-Hop	Post Malone and Swae Lee	"Sunflower"	2	2:38	158	Apple Music
2010	2019	Pop	Halsey	"Without Me"	3	3:22	202	Apple Music
2010	2019	Pop	Billie Eilish	"Bad Guy"	4	3:14	194	Apple Music
2010	2019	Hip-Hop	Post Malone	"Wow"	5	2:30	150	Apple Music
2010	2019	EDM	Marshmello and Bastille	"Happier"	6	3:34	214	Apple Music
2010	2019	Pop	Ariana Grande	"7 Rings"	7	2:59	179	Apple Music
2010	2019	R&B	Khalid	"Talk"	8	3:18	198	Apple Music
2010	2019	Hip-Hop	Travis Scott	"Sicko Mode"	9	5:13	313	Apple Music
2010	2019	Pop	Jonas Brothers	"Sucker"	10	3:01	181	Apple Music
2010	2019	Pop	Panic! at the Disco	"High Hopes"	11	3:11	191	Apple Music
2010	2019	Pop	Ariana Grande	"Thank U, Next"	12	3:27	207	Apple Music
2010	2019	Hip-Hop	Lizzo	"Truth Hurts"	13	2:53	173	Apple Music
2010	2019	R&B	Sam Smith and Normani	"Dancing with a Stranger"	14	2:51	171	Apple Music
2010	2019	Latin	Shawn Mendes and Camila Cabello	"Señorita"	15	3:11	191	Apple Music

Figure 5: 2010s Decade Data Block

	A	B	C	D	E	F	G	H	I
1	DECADE	YEAR	GENRE	ARTIST	SONG TITLE	RANK	LENGTH (MM:SS)	SEC	PLATFORM
2	2020	2021	Pop	Dua Lipa	"Levitating"	1	3:24	204	Apple Music
3	2020	2021	Pop	The Weeknd and Ariana Grande	"Save Your Tears"	2	3:11	191	Apple Music
4	2020	2021	Pop	The Weeknd	"Blinding Lights"	3	3:22	202	Apple Music
5	2020	2021	Hip-Hop	24kGoldn featuring Iann Dior	"Mood"	4	2:21	141	Apple Music
6	2020	2021	Rock	Olivia Rodrigo	"Good 4 U"	5	2:58	178	Apple Music
7	2020	2021	R&B	Doja Cat featuring SZA	"Kiss Me More"	6	3:29	209	Apple Music
8	2020	2021	Soul	Silk Sonic (Bruno Mars and Anderson .Paak)	"Leave the Door Open"	7	4:02	242	Apple Music
9	2020	2021	Pop	Olivia Rodrigo	"Drivers License"	8	4:02	242	Apple Music
10	2020	2021	Hip-Hop	Lil Nas X	"Montero (Call Me by Your Name)"	9	2:50	170	Apple Music
11	2020	2021	R&B	Justin Bieber featuring Daniel Caesar and Giveon	"Peaches"	10	3:18	198	Apple Music
12	2020	2021	Pop	BTS	"Butter"	11	2:44	164	Apple Music
13	2020	2021	Pop	The Kid Laroi and Justin Bieber	"Stay"	12	2:22	142	Apple Music
14	2020	2021	Pop	Olivia Rodrigo	"Deja Vu"	13	3:36	216	Apple Music
15	2020	2021	R&B	Ariana Grande	"Positions"	14	2:52	172	Apple Music
16	2020	2021	EDM	Ed Sheeran	"Bad Habits"	15	3:51	231	Apple Music
17	2020	2020	Pop	The Weeknd	"Blinding Lights"	1	3:22	202	Apple Music
18	2020	2020	Pop	Post Malone	"Circles"	2	3:35	215	Apple Music
19	2020	2020	Hip-Hop	Roddy Ricch	"The Box"	3	3:17	197	Apple Music
20	2020	2020	Disco	Dua Lipa	"Don't Start Now"	4	3:03	183	Apple Music
21	2020	2020	Hip-Hop	DaBaby featuring Roddy Ricch	"Rockstar"	5	3:25	205	Apple Music
22	2020	2020	Pop	Harry Styles	"Adore You"	6	3:27	207	Apple Music
23	2020	2020	Hip-Hop	Future featuring Drake	"Life Is Good"	7	3:58	238	Apple Music
24	2020	2020	Pop	Maroon 5	"Memories"	8	3:09	189	Apple Music
25	2020	2020	Country	Maren Morris	"The Bones"	9	3:17	197	Apple Music
26	2020	2020	Pop	Lewis Capaldi	"Someone You Loved"	10	3:02	182	Apple Music
27	2020	2020	Disco	Doja Cat	"Say So"	11	3:58	238	Apple Music
28	2020	2020	Country	Gabby Barrett featuring Charlie Puth	"I Hope"	12	3:31	211	Apple Music
29	2020	2020	Hip-Hop	Jack Harlow featuring DaBaby, Tory Lanez and Lil Wayne	"Whats Poppin"	13	3:47	227	Apple Music
30	2020	2020	Pop	Tones and I	"Dance Monkey"	14	3:29	209	Apple Music
31	2020	2020	Hip-Hop	Megan Thee Stallion featuring Beyoncé	"Savage"	15	4:02	242	Apple Music
32	2020	2023	Country	Morgan Wallen	"Last Night"	1	2:44	164	Apple Music
33	2020	2023	Pop	Miley Cyrus	"Flowers"	2	3:21	201	Apple Music
34	2020	2023	R&B	SZA	"Kill Bill"	3	2:34	154	Apple Music
35	2020	2023	Pop	Taylor Swift	"Anti-Hero"	4	3:21	201	Apple Music
36	2020	2023	R&B	Metro Boomin, the Weeknd and 21 Savage	"Creepin'"	5	3:53	233	Apple Music
37	2020	2023	Afrobeats	Rema and Selena Gomez	"Calm Down"	6	3:40	220	Apple Music
38	2020	2023	R&B	The Weeknd and Ariana Grande	"Die for You"	7	3:53	233	Apple Music
39	2020	2023	Country	Luke Combs	"Fast Car"	8	4:25	265	Apple Music
40	2020	2023	R&B	SZA	"Snooze"	9	3:22	202	Apple Music
41	2020	2023	EDM	David Guetta and Bebe Rexha	"I'm Good (Blue)"	10	2:55	175	Apple Music
42	2020	2023	Pop	Sam Smith and Kim Petras	"Unholy"	11	2:37	157	Apple Music
43	2020	2023	Country	Morgan Wallen	"You Proof"	12	2:37	157	Apple Music
44	2020	2023	Country	Zach Bryan	"Something in the Orange"	13	4:14	254	Apple Music
45	2020	2023	Hip-Hop	Drake and 21 Savage	"Rich Flex"	14	3:59	239	Apple Music
46	2020	2023	Pop	Harry Styles	"As It Was"	15	2:47	167	Apple Music

Figure 6: 2020s Decade Data Block

Analysis:

We first did a non-randomized blocking analysis in Minitab. Assigning song length in seconds as our response factor. We blocked the data by, decade, genre, and billboard rankings.

Initial analysis considering a 5% significance level presented in Figure 7, showed no significant difference on the mean song lengths in the blocks of 12 genres, or 15 billboard rankings. However, a significant difference in at least one of the means of the decade's blocks is present. The Residual Charts shown in Figure 8, show a largely normal distribution of the residual values at first glance, with a few outliers in the upper 5 percentiles. The outliers appear again in the histogram giving a very slight right skew.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Decade	**5**	**116348**	**23270**	**9.69**	**0.000**
Genre	**11**	**16170**	**1470**	**0.61**	**0.818**
Rank	**14**	**36964**	**2640**	**1.10**	**0.358**
Error	**239**	**573716**	**2400**		
Lack-of-Fit	191	473434	2479	1.19	0.246
Pure Error	48	100283	2089		
Total	**269**	**772955**			

Figure 7: Non-Randomized 270 Song Lengths, Anova Blocking (Decade, Genre, and Billboard Rank)

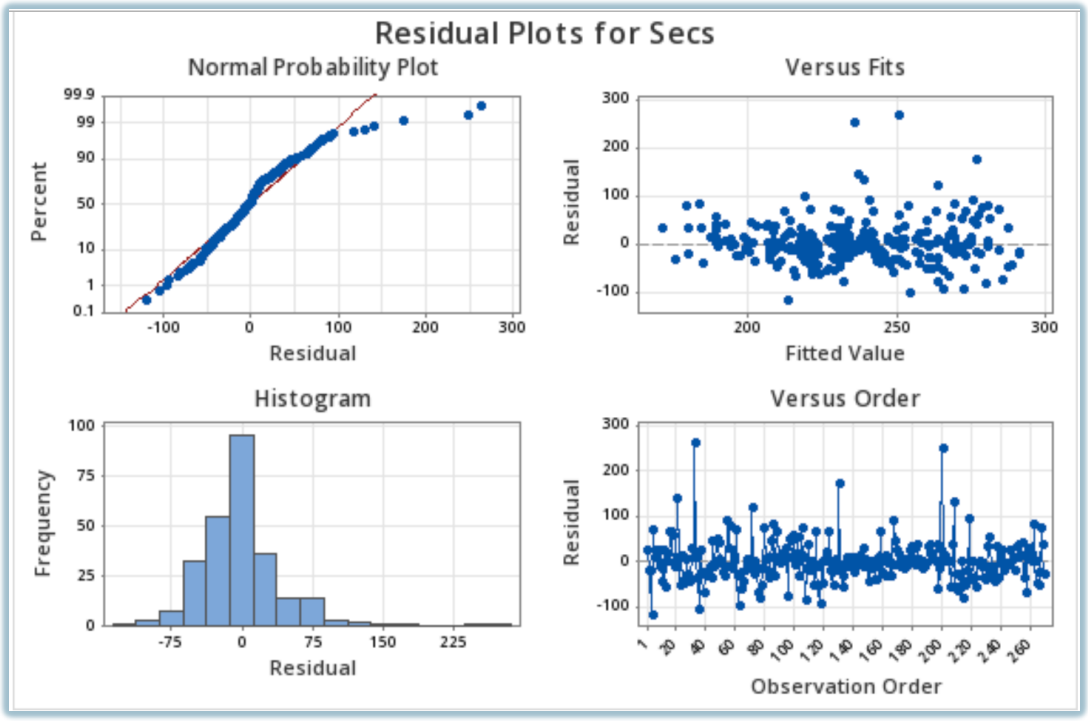


Figure 8: Non-Randomized 270 Song Lengths, Residual Charts

We then performed the same blocking design but with the data set randomized. Presented in Figure 9. The data sampling order had no effect on the data, therefore as expected the only analysis data to change was in the Residual vs Observation order plot, shown in Figure 10.

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Decade	**5**	**116348**	**23270**	**9.69**	**0.000**
Rank	**14**	**36964**	**2640**	**1.10**	**0.358**
Genre	**11**	**16170**	**1470**	**0.61**	**0.818**
- Error	**239**	**573716**	**2400**		
Lack-of-Fit	191	473434	2479	1.19	0.246
Pure Error	48	100283	2089		
Total	**269**	**772955**			

Figure 9: Randomized 270 Song Lengths, Anova Blocking (Decade, Genre, and Billboard Rank)

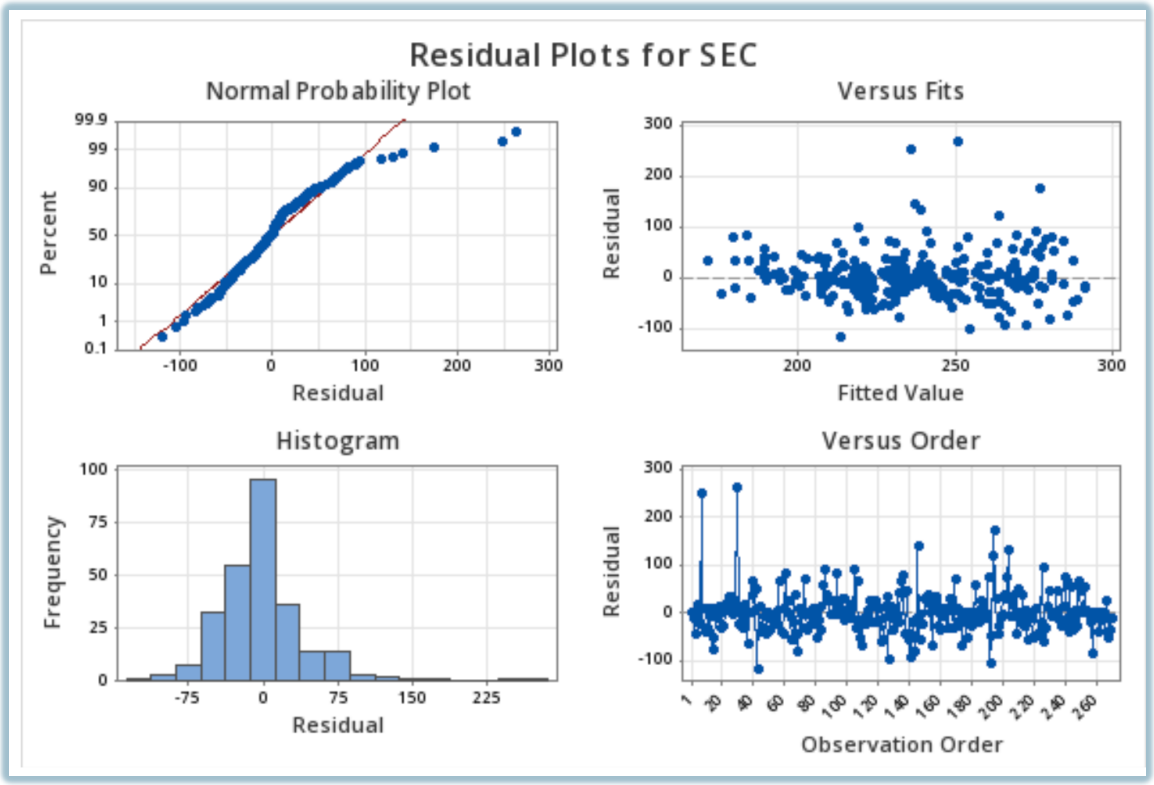


Figure 10: Randomized 270 Song Lengths, Residual Charts

Early analysis of the residuals data validates our model, showing a factor in our model does influence in the song lengths. While the Anova block indicates at least one block sound length mean differs from others.

Next Steps:

We will next analyze the current data while removing each non effective block individually. We will analyze various Tukey Tests to isolate which decades differ. After we will continue to add data to our sample pool and then compare if any trends found continue.